Lakshya JEE 2.0 (2024)

Matrices

DPP-04

1. If A and B are symmetric matrices of the same order, then
(1) AB is a symmetric matrix
(2) A –B is a skew-symmetric matrix
(3) AB + BA is a symmetric matrix
(4) AB – BA is a symmetric matrix

2. If $A = \begin{vmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}$, then the value of |adj A|, is
(1) a^{27} (2) a^9
(3) a^6 (4) a^2

3. If A is a 3 × 3 matrix and B is its adjoint such that $|B| = 64$, then $|A| =$
(1) 64 (2) ± 64
(3) ± 8 (4) 18

4. If the adjoint of a 3 × 3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible values of the determinant of P are
(1) ± 2 (2) ± 1
(3) ± 3 (4) ± 4

5. If $A = [a_{ij}]_{4 \times 4}$ such that $a_{ij} = \begin{cases} 2, & i = j \\ 0, & i \neq j \end{cases}$,
Then $\left\{\frac{1}{7}\det\left(\text{adj}(\text{adj}A)\right)\right\}$ is, where {.} denotes the fractional part function.
(1) 1/7 (2) 2/7
(3) 3/7 (4) None of these

6. If A is a singular matrix, then adj A is
(1) non-singular
(2) singular
(3) symmetric
(4) not defined

7. If the matrices $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, B = adj A and $C = 3A$, then $\frac{|\text{adj}B|}{|C|}$ is equal to
(1) 72 (2) 8
(3) 16 (4) 2

8. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$, then adj(adj A) is:
(1) $\begin{bmatrix} 3 & 3 & 3 \\ 6 & 9 & 15 \\ 9 & 15 & 36 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$
(3) $\begin{bmatrix} 3 & 6 & 9 \\ 3 & 9 & 15 \\ 3 & 15 & 36 \end{bmatrix}$ (4) None of these

9. If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then the $\frac{1}{1000}$ [|maximum value of Δ – minimum value of $\Delta|^3$] is equal to ____.
(1) 1 (2) –1
(3) 0 (4) 2

10. If the value of a third order determinant is 11, then value of the square of the determinant formed by the cofactors will be
(1) 11 (2) 121
(3) 1331 (4) 14641

11. If A and B are square matrices of order 3 such that $|A| = -1$, $|B| = 3$, then $|3\,AB|$ equals
(1) – 9 (2) – 81
(3) –27 (4) 81

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (3)
3. (3)
4. (1)
5. (1)
6. (2)
7. (2)
8. (3)
9. (1)
10. (4)
11. (2)